

Performance Testing

Project: ComicCraft | AI/ML & Generative AI

1. Testing Objective

Performance testing evaluates the responsiveness and stability of the ComicCraft application during normal use and repeated requests. The focus is on application startup, frontend loading, backend response handling and Generative AI response time.

2. Test Areas

Test Area	Test Description	Expected Result
Application Startup	Start the FastAPI application and observe startup behavior.	Application starts without avoidable errors.
Frontend Loading	Open the ComicCraft web page in a modern browser.	Interface loads correctly and is usable.
API Request	Submit a valid comic prompt to the backend.	Request is processed and a response is returned.
AI Generation	Measure the time between prompt submission and Gemini AI response.	Generated content is returned successfully.
Repeated Requests	Submit multiple valid prompts during a test session.	Application continues to respond without unexpected failure.
Error Handling	Test invalid input or unavailable AI configuration.	Clear error feedback is shown to the user.

3. Performance Factors

- Network latency between the application and Generative AI service.
- Gemini AI response time and generated output size.
- Backend processing time.
- Frontend rendering and display time.
- Number of requests made during testing.

4. Performance Improvement Opportunities

- Validate prompt input before calling the AI service.
- Avoid unnecessary API requests from the frontend.
- Provide loading feedback while generation is in progress.
- Handle failures without blocking the complete application.
- Add monitoring and rate limiting for larger deployments.

5. Measurement Record

Exact response-time measurements should be recorded during final testing because results depend on the local machine, network conditions and the AI service at the time of testing.

6. Test Conclusion

The performance test plan covers the critical user journey from opening ComicCraft to receiving generated comic content. The same test cases can be repeated after future image-generation features are added.